

Improving Autonomous Vehicle Cognitive Robustness in Extreme Weather With Deep Learning and Thermal Camera Fusion

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ABSTRACT In autonomous vehicles (AV), sensor fusion methods have proven to be effective in merging data from multiple sensors and enhancing their perception capabilities. In the context of sensor fusion, the distinct strengths of multi-sensors, such as LiDAR, RGB, Thermal sensors, etc., can be leveraged to mitigate the impact of challenges imposed by extreme weather conditions. In this paper, we address multi-sensor fusion in AVs and present a comprehensive integration of a thermal sensor aimed at enhancing the cognitive robustness of AVs. Thermal sensors possess an impressive capability to detect objects and hazards that may be imperceptible to traditional visible light sensors. When integrated with RGB and LiDAR sensors, the thermal sensor becomes highly beneficial for detecting and locating objects in adverse weather conditions. The proposed deep learning-assisted multi-sensor fusion technique consists of two parts: (1) visual information fusion and (2) object detection using LiDAR, RGB, and Thermal sensors. The visual fusion framework employs a CNN (convolutional neural network) inspired by a domain image fusion algorithm. The object detection framework uses the modified version of the YoloV8 model, which exhibits high accuracy in real-time detection. In the YoloV8 model, we adjusted the network architecture to incorporate additional convolutional layers and altered the loss function to enhance detection accuracy in foggy and rainy conditions. The proposed technique is effective and adaptable in challenging conditions, such as night or dark mode, smoke, and heavy rain. The experimental results of the proposed method demonstrate enhanced efficiency and cognitive robustness compared to state-of-the-art fusion and detection techniques. This is evident from tests conducted on two public datasets (FLIR and TarDAL) and one private dataset (CUHK).

INDEX TERMS Sensor fusion, Thermal camera, Object detection, Object recognition, convolution neural network.

NOMENCLATURE

AV	Autonomous vehicles.
LiDAR	Light detection and ranging.
RGB	Red, green blue.
AI	Artificial intelligence.
CNN	Convolutional neural network.
CUHK	Chinese University of Hong Kong.
SAE	Society of automotive engineers.
MRE	Mean re-projection error.
HKPC	Hong Kong productivity council.
CVAT	Computer vision annotation tool.
TP	True positive.
FN	False negative.
FP	False positive.

I. INTRODUCTION

Sensor fusion is a fundamental aspect of autonomous vehicle technology, which encompasses the merging of data from multiple sensors to create a comprehensive representation of the vehicle's surroundings. In the context of AV, sensor fusion commonly involves the integration of information from a variety of sensors, such as radar, thermal cameras, LiDAR, and RGB cameras. By using a combination of these sensors, AVs can navigate more safely and efficiently.

The SAE J3016 [1] classification divides AV technology into six levels of automation, ranging from full automation (Level 5) to no automation (Level 0), as shown in Fig. 1. AVs have the potential to bring numerous benefits, including increased safety, better fuel economy, and reduced traffic congestion [2]. However, there are still numerous issues that need to be resolved before driverless vehicles become a common sight on the roads. So, there have been many studies that have explored the use of different sensors for sensor fusion in autonomous vehicles [2], [3], [4]. Research has shown that sensor fusion can enhance the precision and reliability of object recognition and localization while improving the efficiency and safety of AVs.

Over the past decade, various traditional [5], [6] and deep learning [7], [8], [9], [10], [11], [12] techniques have been proposed to fuse different sensor domains. Traditionally, sensor fusion techniques have relied on handcrafted algorithms to process sensor data and perform fusion [6]. However, with the advent of deep learning techniques, there has been a paradigm shift towards the use of neural networks for sensor fusion in AVs [10], [11]. Deep learning-based sensor fusion methods offer several advantages over traditional approaches. They learn to get relevant patterns and features immediately from the sensor data, enabling the system to adapt and generalize to different driving scenarios.

Recently, various deep learning architectures and algorithms have been studied to improve the efficiency and reliability of sensor fusion in AVs. Deep learning systems can efficiently manage large amounts of data, enabling more accurate and reliable sensor fusion.

One of the primary benefits of sensor fusion is that multiple sensors can compensate for the shortcomings of individual

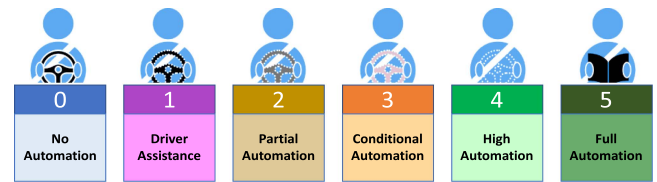


FIGURE 1. Illustrates the different levels of autonomous vehicles.

Tasks	Camera	Thermal	LiDAR	Fusion
Object detection	+	+	+	✓
Pedestrian detection	+	+	+	✓
Weather condition	-	+	○	✓
Lighting condition	-	+	+	✓
Dirt	-	+	○	✓
Velocity	○	○	○	✓
Distance accuracy	○	-	+	✓
Distance range	○	○	+	✓
Data density	+	+	-	✓
Classification	○	○	+	✓
Packaging	○	+	○	✓

Strength (+) Capability (○) Weakness (-) Effective (✓)

FIGURE 2. Illustrates the strength of different sensors in different conditions.

sensors. For example, high-resolution photos from RGB cameras can be affected by weather and lighting conditions. In contrast, thermal cameras are not affected by sunlight and can detect infrared radiation emitted by objects. While LiDAR sensors cannot see through dense fog or heavy rain, they can create high-resolution 3D models of the vehicle's surroundings. Radar sensors can detect objects in bad weather but have lower resolution compared to LiDAR. By combining information from all these sensors, an AV can create a more accurate picture of its environment.

Reducing the possibility of false alarms or incorrect detection is another benefit of sensor fusion. A single sensor might generate false alarms or incorrectly identify an item due to noise, interference, being out of focus, or being obscured by another object. By integrating data from several sensors, an autonomous car can lessen the likelihood of these issues.

The capabilities of LiDAR, thermal cameras, and RGB cameras vary (Fig. 2) [3]. Merging their data enhances our overall view of the surroundings. For example, RGB cameras enable the vehicle to identify and detect objects based on color and shape, while thermal cameras allow identification based on temperature. LiDAR provides a 3D point cloud of the surroundings, helping the vehicle understand the shape and distance of nearby objects. Sensor fusion, which combines data from LiDAR, thermal cameras, and RGB cameras, is one of the main technologies in AVs. In addition to compensating for the limitations of individual sensors, this technology increases vehicle decision-making accuracy and safety. Overall,

it improves the vehicle's perception of its environment. By using factors like color, temperature, shape, and distance, the vehicle can identify and detect objects more effectively.

Contributions: The main contributions of our work can be summarized as follows:

- We proposed an approach that uses a Convolutional Neural Network (CNN) to fuse data from multiple sensors (Thermal, RGB, and LiDAR) in AVs to enhance the accuracy of detecting surrounding objects.
- We used a thermal camera to detect and recognize objects in adverse weather conditions such as smoke, fog, dust, fire, and heavy rainfall, during which RGB and LiDAR sensors are not functioning optimally.
- We applied modified Yolov8 and PointPillars frameworks to create a real-time object detection system for AVs. These models efficiently detect the surrounding objects and can also determine the speed of the vehicles using other sensors (Radar, Ultrasonic, etc.).
- We conducted experiments on public and private datasets to validate the effectiveness of our sensor fusion and object detection models. Our results demonstrated a significant increase in both quantity and quality compared to existing sensor fusion approaches.

The rest of the paper is organized as follows: In Section II, we review relevant research on sensor fusion for object recognition, using various machine learning and deep learning approaches. Section III presents a proposed sensor fusion framework for object detection. Section IV details the evaluation process and experimental findings. In Section V, we compare models and discuss future work and limitations. Finally, Section VI concludes the overall success of the proposed technique.

II. RELATED WORKS

In the field of autonomous cars, sensor fusion is a widely researched topic. Numerous studies have explored the use of various sensors, including LiDAR, radar, thermal cameras, and RGB cameras, for sensor fusion. One of the most popular techniques for autonomous cars involves using RGB cameras and LiDAR [13]. This combination of sensors has proven effective for localization and object detection. Zheng et al. [4] presented a sensor fusion system for autonomous driving in urban settings that utilizes RGB cameras and LiDAR. This technology enables real-time detection and tracking of automobiles, pedestrians, and bicycles.

An alternative method for sensor fusion in autonomous cars is the use of thermal cameras [14]. As shown in Fig. 3, these cameras are not affected by lighting conditions and can detect infrared radiation emitted by objects. This capability makes them ideal for use in bad weather, such as heavy rain or fog. Wang and colleagues [15] developed a sensor fusion system for autonomous driving in such conditions, combining thermal cameras and LiDAR. Their system successfully detected and tracked both pedestrians and moving cars in heavy rain and fog.

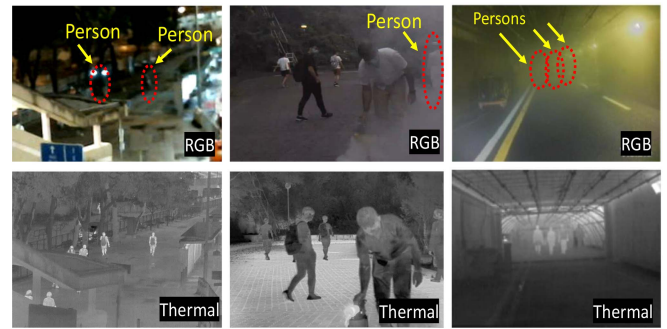


FIGURE 3. Shows the strength of a thermal camera during the night and smoke weather in the same scene. The first row shows the RGB frames with low visibility of objects due to the smoke and darkness, while the second row portrays the thermal sensor frames with clear visibility of objects in the smoke and night mode (best view in color).

Furthermore, Hashem et al. [8] presented a multi-sensor fusion technique based on a probabilistic data association method. This approach uses information from four main sensors radar, LiDAR, an RGB camera, and an infrared camera to improve detection accuracy. Recommendations for object zones are generated based on the fused detection findings. However, because it is largely focused on the sea environment, this technique has not been thoroughly evaluated in real-world traffic situations. In another study, Dong et al. [14] described a sensor fusion system that integrates a LiDAR sensor with a thermal infrared camera for object identification and recognition. A 3D calibration target is created and used in their proposed method to acquire the exterior parameters of both sensors. However, this technique is not very effective during heavy rain or snowfall.

Radar sensors have been used alongside other sensors, including cameras and LiDAR, to enhance the perception of the surrounding environment in various research studies. For example, Zhang et al. [16] proposed a sensor fusion system that utilizes cameras, radar, and LiDAR in their research. This technology enabled real-time detection and tracking of automobiles, pedestrians, and bicycles. Additionally, some studies have suggested using ultrasonic sensors for sensor fusion in self-driving cars. Ultrasonic sensors are suitable for close-range applications and can provide accurate distance measurements. Li et al. [17] research suggested using a sensor fusion system for autonomous parking that included cameras and ultrasonic sensors. This technology enabled real-time tracking of parking spaces and vehicles.

By integrating these sensors, the autonomous car can better perceive its environment, which enhances item detection, location accuracy, and reliability. For example, Jung et al. [18] conducted research and proposed a sensor fusion system for autonomous driving in urban areas. This system included LiDAR, radar, RGB cameras, and ultrasonic sensors. The technology performed well in tracking automobiles, pedestrians, and bicycles. Therefore, various fusion methodologies show that the more sensors you have, the more accurate the fusion models output will be.

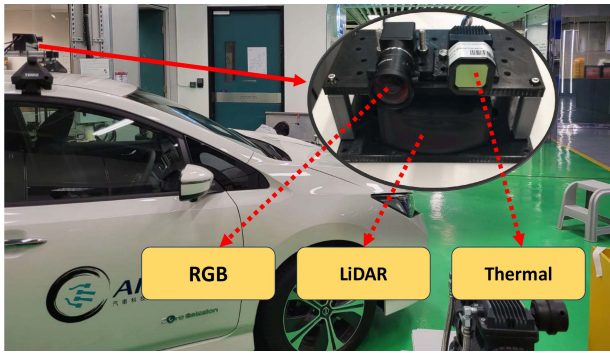


FIGURE 4. Shows the sensor array mounted on the top of the AV.

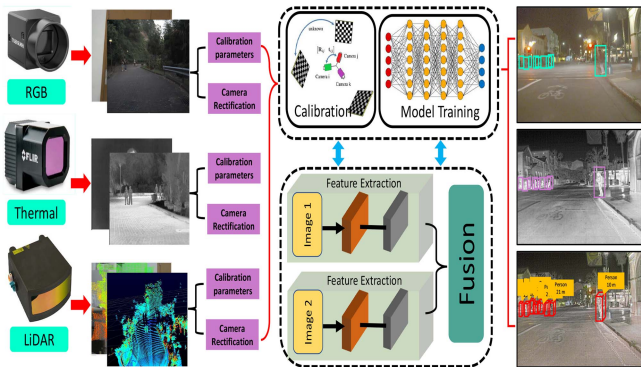


FIGURE 5. The flow chart for the proposed technique is divided into three sections. The first section focuses on sensor data gathering and calibration. The second section involves sensor fusion and training. Finally, the third section is dedicated to object recognition (best view in color).

III. PROPOSED FRAMEWORK

A. SENSORS DATA ACQUISITION AND ENVIRONMENT

The proposed sensor fusion framework uses three types of sensors, which are mounted on the top of the AV, as shown in Fig. 4. These sensors are powered either by the AV power system or by a portable AC power bank. Details for each sensor are provided below.

1) RGB CAMERA

To produce the high-resolution image, as shown in Fig. 5, we utilized the Sony IMX250 sensor [19], which is integrated into the MV-CH050-10UM/UC camera. This sensor allows for the transmission of real-time, uncompressed images through a USB 3.0 interface, with a maximum frame rate of 74 frames per second. Additionally, the camera offers automatic and manual adjustments for gain, exposure control, LUT, gamma correction, and other settings. It is equipped with M2 screw holes to secure the cable connecting the USB 3.0 interface and is compatible with USB-3 Vision Protocol, GenlCam Standard, and third-party software.

2) THERMAL CAMERA

The FLIR ADK [20] is the world's most innovative thermal camera core that offers the lightest, smallest, and least

power-consuming high-performance uncooled thermal imaging technique, as shown in Fig. 5. It is used to create devices for firefighting, automotive, marine, hunting, and military applications. Its Boson architecture not only increases the camera's computational capabilities but also drastically reduces the cost of production. The thermal cameras of today enable fine luxury automakers to detect people and vehicles on the roadway. The FLIR ADK is making thermal technology more widely accessible to a greater range of vehicles.

3) LiDAR

In order to gather 3D point data from diverse landscapes, we opted for the RS-LiDAR-M1 sensor [21]. Developed by RoboSense exclusively for mass-produced automobiles, this automotive-grade solid-state LiDAR provides highly precise perception of the 3D environment, ensuring safe driving. The M1 boasts several benefits, including its durability, affordability, ease of mass production, and seamless integration with the vehicle's body. We collected data from scenes containing various objects, such as animals, humans, vehicles, and pedestrians. We are collecting data for two weather conditions for each sensor, as discussed below.

4) SURROUNDING ENVIRONMENTS

We gathered data from two distinct environments at various locations throughout Hong Kong. The collected data consisted of various classes: people, cars, bicycles, motorcycles, buses, trucks, traffic lights, other vehicles, etc.

Normal weather: Here normal weather means no rain, fog, smoke, dust, etc. We have captured data at three different times (day, evening, and night) in normal weather.

Adverse weather: It contains smoke, dust, and rain. We generated an artificial smoke-weather scenario to capture the data, as shown in the second row of Fig. 3. We have captured rainy weather data by using RGB, thermal, and LiDAR sensors in various locations in Hong Kong, such as Tai Wai Bridge, Science Park, and CUHK Hong Kong.

5) SENSORS STRENGTH

The idea of a driverless vehicle has sparked the interest of many people, leading some to doubt that such a possibility even exists or will exist soon. Fig. 2 illustrates the strength of the sensor in different weather. So, developing an artificial intelligence driving system that can control the vehicle's parts in real-time is an incredibly difficult task in AV. The biggest challenge lies in programming the AV system to handle the vehicle's performance in any given situation using hundreds of sensors located throughout the vehicle, which is a complex task. Thermal sensors are effective in adverse weather, high lighting, and night mode, whereas RGB sensors are ideal for object detection and classification. Meanwhile, LiDAR is highly efficient for pedestrian detection, velocity, 360° surroundings object detection, distance accuracy, data density,

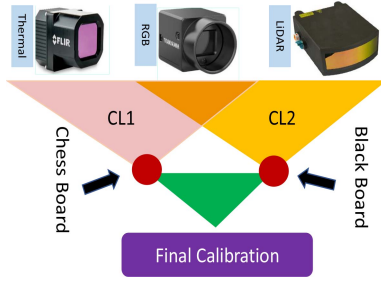


FIGURE 6. Shows the sensor calibration of three sensors. We divide it into two parts: (1) RGB and thermal calibration, and (2) RGB and LiDAR calibration.

classification, and dirt identification. The combination of sensors' strengths can significantly enhance the performance of the AV system.

B. SENSORS CALIBRATION

To remove distortion and aberrations between sensor domains, we employed a sensor calibration model that utilized three sensors: RGB, Thermal, and LiDAR. Algorithm 1 shows the detailed structure of the sensor calibration model. The sensor calibration process is divided into two types of calibrations as shown in Fig. 6: (1) RGB and Thermal sensors calibration (CL_1), and (2) RGB and LiDAR sensors calibration (CL_2). So, the final calibration is the combination of CL_1 and CL_2 , which is defined as;

$$CL_f = CL_1 \oplus CL_2, \quad (1)$$

1) RGB AND THERMAL CALIBRATION

Geometric calibration is a critical process for multi-view geometry-based applications, as it ensures an accurate 3D-to-2D mapping between 3D object points and 2D image points. This is especially important because the optical distortion makes the image deformable. So, the calibration helps to ensure precise data capture for the purposes of better algorithm and application accuracy and performance. We used distortion-free projective transformation given by the pinhole camera model [22] for CL_1 calibration.

$$CL_1 = s * p = A * [R|t] * P_w \quad (2)$$

where a 2D pixel p shows the image plane. P_w shows the world coordinate system. A camera intrinsic matrix A , a rotation matrix R , and a translation matrix t describing the change of coordinates from world to camera (or camera frame) coordinates. In projective transformation, an arbitrary scaling factor (s) does not form a part of the camera model.

$$p = A * P_c \quad (3)$$

$$A = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$s * \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \quad (5)$$

The combined rotation-translation matrix $[R|T]$ results from both a homogeneous and projective transformation. The projective matrix is presented by 3-by-4 transformation maps in camera coordinates 2D points where $X = X/Z$, and $Y = Y/Z$.

$$Z_c * \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} * \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} \quad (6)$$

$$P_c = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} * P_w, \quad (7)$$

$$\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (8)$$

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (9)$$

After combining projectile transformation and homogeneous transformation, we get the required transformation matrix in world coordinates 2D points.

$$Z_c * \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = [R|t] * \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} * \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (10)$$

with $x = X_c/Z_c$, and $y = Y_c/Z_c$. So, we get the final calibration (1) by combining intrinsic and extrinsic parameters together.

$$s * \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} * \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (11)$$

We used the chessboard for the calibration experiment according to Zhang's approach [23], which was implemented in the OpenCV library. This enabled the detection of the chessboard pattern in both RGB and thermal cameras, as shown in Fig. 7 (a & b). The thermal camera produced heat patterns when exposed to sunlight because the black area absorbs more heat than the white area, as shown in Fig. 7(b). To measure the quality of the calibration, we compared the Mean Reprojection Error (MRE) between thermal and RGB cameras with the standard chessboard target. The use of the chessboard

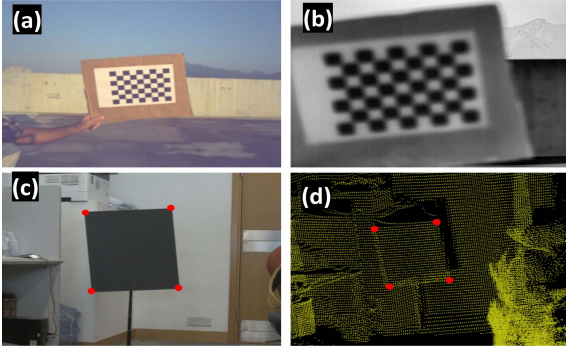


FIGURE 7. Shows the chessboard and black board experiment for calibration. (a) and (b) shows the RGB and thermal sensor frame of the same scene, respectively. (c) and (d) are RGB and LiDAR frames of the same scene, respectively.

Algorithm 1: Sensors Calibration.

Input : RGB frames $\{I_{RGB}\}$, Thermal frames $\{I_T\}$,
LiDAR point cloud $\{P_{lidar}\}$, and parameter (P) .

- 1: **if** parameter $P = 1$,
- 2: Calibrate $CL_1 = \{I_{RGB}, I_T\}$,
- 3: **input** $\rightarrow \{Images(I_{RGB}, I_T), Camera\ param. (C_p)\}$,
- 4: Calibration using OpenCV: $\{I_{RGB} \odot I_T\}$,
- 5: **output** $\leftarrow calibrated(I_{RGB}), calibrated(I_T)$
- 6: **if** parameter $P = 2$,
- 7: Calibrate $CL_2 = \{I_{RGB}, P_{lidar}\}$,
- 8: **input** $\rightarrow \left\{ \begin{array}{l} Images(I_{RGB}), Point\ cloud(P_{lidar}), \\ Camera\ parameters(C_p) \end{array} \right\}$
- 9: Calibration using Livox/OpenCV: $\{I_{RGB} \odot P_{lidar}\}$
- 10: **output** $\leftarrow calibrated(I_{RGB}), calibrated(P_{lidar})$

Final Output: $\left\{ \begin{array}{l} Calibrated(I_{RGB}, I_T, P_{lidar}), \\ Extrinsic\ parameters \end{array} \right\}$

calibration technique facilitated thermal geometric calibration while producing high-quality, and accurate results.

We captured RGB and thermal data from distances of 5 to 20 meters at different angles. We converted these images to grayscale using an OpenCV function, which identified the boundaries of the blocks in the frame and enabled us to accurately calculate their position and angles. The extrinsic and intrinsic parameters for calibration between RGB and Thermal frames are obtained from the block size and angles.

2) RGB CAMERA AND LIDAR CALIBRATION

In this step, the Livox method [24] was employed to calibrate the RGB camera and LiDAR sensor using their respective calibration parameters. This approach is implemented in C++, which records a ROS bag file with three sensors (Thermal, RGB, and LiDAR). Fig. 7 (c & d) presents the results of RGB and LiDAR point cloud calibration. For RGB camera-LiDAR calibration (CL_2), we calibrate the 2D image points coordinates $K = [x_1, x_2, \dots, x_n]$ and corresponding 3D coordinates points of LiDAR frame $L = [P_1, P_2, \dots, P_n]$. We perform rotation and transformation T_L by minimizing the reprojection

error of all 2D key points using the Levenberg Marquat optimizer [25].

$$T_L = \arg T_L \min \sum_{j=1}^M p(||\pi(T_L * P_j^K) - x_j^K||^2), \quad (12)$$

we calculate the intrinsic and distortion correct parameters from the image and LiDAR points using the Rosbag function [26], [27].

$$P_i^L = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}, \quad (13)$$

$$D_L = [k_1, k_2, p_1, p_2, k_3], \quad (14)$$

where f_x, c_x, f_y , and c_y are four coordinates positions of blackboard LiDAR points, as shown in Fig. 7(d). The D_L is the distortion correction parameters, where k_1, k_2 are radial distortion parameters; p_1, p_2 are tangential distortion, and k_3 set to 0 by default. The extracted LiDAR edges will match the corresponding edges in images, as shown in Fig. 7 (c & d). Suppose each sample point of LiDAR $P_i^L \in R^3$ is transformed into camera frame $T_i^C = (R_L^C, t_L^C) \in R^2$. So, the RGB camera frame points are transformed as;

$$P_i^C = T_i^C(P_i^L) \in R^3 \quad (15)$$

where $T_i^C(P_i^L) = R_L^C(P_i^L + t_L^C)$ shows the rigid tranformation T_L^C to point P_i^C . The transformed point P_i^C is projected onto the camera image plane to produce the location of the expected points.

$$P_i^C = \pi(P_i^C) \quad (16)$$

where $\pi(P_i^C)$ is the pinhole camera model [22], [28]. We utilized the intrinsic calibration parameters from the RGB sensor and corner point positions of the LiDAR and RGB frames to calculate the extrinsic parameters for the final calibration. We employed an OpenCV-based stereo calibration method to calibrate the RGB and LiDAR sensors. Then, we used the intrinsic and extrinsic parameters of the thermal, RGB, and LiDAR sensors for final calibration (CL_f), as discussed in (1).

C. SENSORS FUSION, OBJECT DETECTION, AND MODEL IMPLEMENTATION

1) SENSORS FUSION

There are three ways to combine the inputs of three sensors: early, late, and hybrid fusion. (1) Early Fusion combines inputs in the early stages (raw inputs or raw features). It needs perfect alignment and synchronization among the sensors to accurately detect objects in the surroundings of the AV. If the data do not match, it can lead to errors. It also takes a long processing time. (2) Late Fusion processes each sensor separately and combines the results in the end. Each sensor processing and combination takes a long time to detect real-time objects. It is not robust if the data is noisy or misaligned. (3) Hybrid Fusion fuses the two approaches by integrating feature levels and sensor data. It extracts the details of the

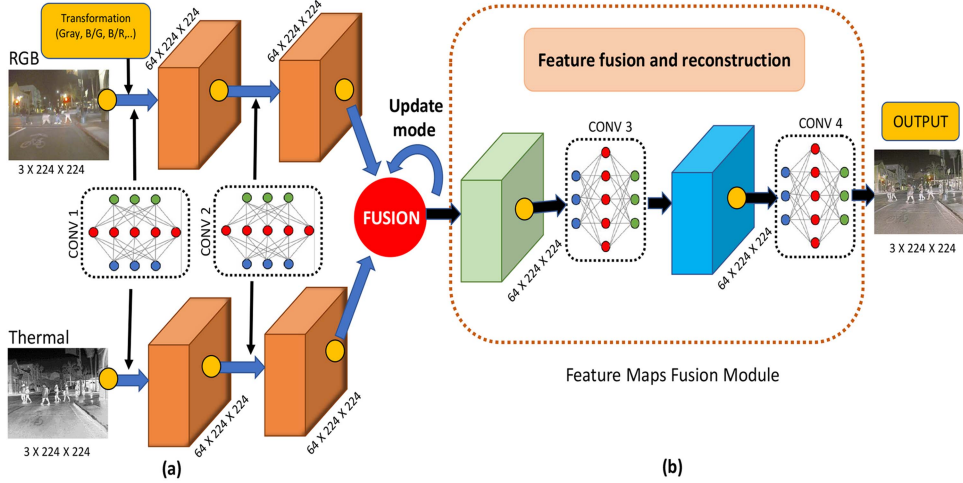


FIGURE 8. Illustrates the deep learning-based fusion model. It consists of two types of modules (a) feature extraction and (2) features map fusion and reconstruction module.

object features in real-time which helps to better understand the object detection process. It is robust as compared to early and late fusion. Algorithm 2 shows the detailed structure of the sensor fusion process. We used a deep learning model [29] to fuse the RGB and thermal sensor images, as shown in Fig. 8. This model usually contains two sub-modules to fuse the high-level features of multiple inputs: (1) multiple feature extraction and concatenation along channel dimensions; and (2) fusing final maps by using element-wise fusion rules. So, the element-wise mean fusion rule is defined as:

$$F(x, y) = fuse(f_i^j, c_2(x, y)), 1 \leq i \leq N \quad (17)$$

where F denotes the future map obtained by our model. The $fuse$ shows the element-wise fusion rule, and the f_i^j, c_2 represents the j^{th} feature frame of the convolution layer. In the first module (Fig. 8(a)), we extract high-level features without down-sampling the feature maps from convolutional layers. There are two convolutional layers in the first module. The size of the first convolution layer (CONV1) is 7×7 , and the size of the second convolution layer (CONV2) is 3×3 , respectively. In the second module (Fig. 8(b)), the kernel number and kernel size of the first convolution layer (CONV3) are set to 64 and 3×3 , respectively. The second convolution layer (CONV4) in the second module is often implemented by the element-wise weighted average [30]. So, the kernel number and kernel size in the second convolution layer of the second module are set to 3 and 1×1 , respectively.

To overcome the over-fitting problem and for boosting purposes, we apply the ReLU activation layer [31] and batch normalization [32] on both middle layers (CONV2, CONV3). Our approach is designed to combine multiple RGB and Thermal images into a single merged image. Additionally, adapting this model to merge single-channel thermal and three channels RGB images is a straightforward process. This model is quick and efficient in combining these images, as illustrated in Fig. 8.

Algorithm 2: Fusion and Object Detection.

Input : RGB frames $\{I_{RGB}\}$, Thermal frames $\{I_T\}$, LiDAR point cloud $\{P_{lidar}\}$, Deep learning model $\{DLM\}$, Fused Images, and Yolo Model (YM).

- 1: **if** loop $L = 1$,
- 2: **input** $\rightarrow \{Images(I_{RGB}, I_T), Camera\ param. (C_p)\}$,
- 3: Fusion using $DLM = \left\{ \begin{matrix} Images(I_{RGB}, I_T), Camera \\ param. (C_p) \end{matrix} \right\}$
- 4: **out** $\leftarrow Fused\ image \leftarrow FUSION$
- 5: **if** loop $L=2$,
- 6: **input** $\rightarrow \{Images(I_{RGB}, I_T), Camera\ param. (C_p)\}$,
- 7: Training $YM = \{Images(I_{RGB}, I_T, P_{lidar}), YM\}$,
- 8: Data distribution $= \{Training, Testing, Validation\}$
- 9: Detection box: $\{I_T \odot I_{RGB}\} || \{I_{RGB} \odot P_{lidar}\}$
- 10: **out** $\leftarrow$ Weights : $\{W_T, W_{RGB}, W_{lidar}\}$, Scores
- 11: **if** loop $L=3$,
- 12: **input** $\rightarrow \{W_{THR}, W_{RGB}, W_{lidar}, Fused\ Images\}$,
- 13: Detection $= \{(2D + 3D\ detection) \diamond Fused\ Images\}$
- 14: **out** $\leftarrow \left\{ \begin{matrix} Fused\ Images, Detection\ Boxes, \\ Distance\ Values \end{matrix} \right\}$

Output: Fusion frames, object detection, and AV relative distance

Ultimately, the following procedures lead to the merging of RGB and thermal images; (1) the input RGB images are transformed into the $YCbCr$ color space, after which the $YCbCr$ channels are separated. (2) each source image multi-channel is then fused by our image fusion model to produce a single Y channel. (3) the Cb and Cr channels of the source images are fused using the same weighted approach described in Prabhakar et al. [33]. Finally, the fusion image is created by transforming the Y channel, the fused Cb channel, and the fused Cr channel back to the RGB color space. Thus, it is necessary to convert the Y channel of each source picture into

TABLE 1. Shows the Model Parameters of YOLOv8 and PointPillar Used in the Proposed Model

Parameters	YOLOv8 Model	PointPillar Model
Backbone	CSP-based architecture	Pillar feature encoding (PFE)
Input Resolution	Adjustable (e.g., 640x640)	Fixed, typically 3D lidar point cloud
Detection Type	2D Object Detection (images/videos)	3D Object Detection (point clouds)
Anchor Boxes	Anchor-free design	Anchor-based design
Output Format	Bounding boxes (x, y, w, h, confidence)	3D bounding boxes (x, y, z, l, w, h, yaw)
Optimization	Focuses on real-time performance and accuracy	Optimized for sparse 3D data processing
Use Case	General-purpose object detection	Autonomous driving and robotics
Pretrained Models	Available for common datasets like COCO	Requires training on 3D datasets (e.g., KITTI)
Framework	PyTorch	TensorFlow or PyTorch

TABLE 2. Quantitative Comparison of Different Sensor Fusion Methods on *FLIR* Dataset [47]

Metrics	GFF	MECNN	LPSR	MFCNN	MFIF	Proposed
VIF [55]	0.6806	0.5606	0.6830	0.6804	0.7823	0.7826
SSIM [56]	0.6187	0.5463	0.6178	0.6206	0.6418	0.6493
MSSIM [57]	0.6997	0.6178	0.6987	0.7018	0.7258	0.7343
MSE [57]	19.213	19.312	18.012	17.860	17.430	15.405
RMSE [57]	4.3832	4.3941	4.2440	4.2261	4.1749	3.9249
NMI [58]	1.0440	0.8436	0.9816	0.8435	0.9034	0.9056
SF [59]	19.280	7.8870	19.410	19.203	19.430	19.433
AG [60]	2.8650	1.1942	2.8891	2.8534	2.8870	2.8730

TABLE 3. Quantitative Comparison of Different Sensor Fusion Methods on the *TarDAL* Dataset [48]

Metrics	GFF	MECNN	LPSR	MFCNN	MFIF	Proposed
VIF [55]	0.557	0.3168	0.7328	0.7760	0.6625	0.7589
SSIM [56]	0.4272	0.4495	0.4465	0.6185	0.6352	0.6989
MSSIM [57]	0.4831	0.5083	0.5049	0.6995	0.7185	0.7905
MSE [57]	19.708	17.912	17.441	12.675	12.603	12.040
RMSE [57]	4.4393	4.2322	4.1762	3.5601	3.5500	3.4698
NMI [58]	0.3614	0.3785	0.3652	0.3985	0.3799	1.0470
SF [59]	9.6250	6.2453	10.482	9.5032	12.256	12.312
AG [60]	1.5062	0.9498	1.6892	1.3589	1.7810	1.8920

three channels (R, G, and B) when feeding RGB and Thermal images into our image fusion model.

$$Y^* = 0.299 \times R + 0.587 \times G + 0.114 \times B, \quad (18)$$

Where R , G , and B are the three channels of the fused image, and 0.299, 0.587, and 0.144 are optimized parameter values [34] in the red, green, and blue channels, respectively. We tested the fusion model for single and multiple objects. The results have shown an 85% similarity index value in sensor fusion results (Tables 2, 3, and 4). We conducted experiments during the day and night to check the stability of the proposed fusion model.

2) OBJECT DETECTION (THERMAL, RGB, LIDAR)

We used two types of deep learning models to train the thermal, RGB, and LiDAR data for object detection. The RGB and Thermal data are trained on the modified YoloV8 model [35], [37], and LiDAR data is trained on the PointPillar model [36]. The parameters detail of both models is given in Table 1. These are pre-trained models, which are used to detect the surrounding objects in AVs. Fig. 9 shows a detailed overview of the object detection model training. Three portions were taken from the total number of images and point cloud frames: 80% for training, 10% for validation, and 10% for testing.

TABLE 4. Quantitative Comparison of Different Sensor Fusion Methods on the CUHK Dataset

Metrics	GFF	MECNN	LPSR	MFCNN	MFIF	Proposed
VIF [55]	0.6714	0.5223	0.7782	0.7925	0.6799	0.7989
SSIM [56]	0.4150	0.3275	0.4778	0.5536	0.7536	0.7895
MSSIM [57]	0.4693	0.3704	0.5403	0.6261	0.8523	0.8939
MSE [57]	23.265	22.350	16.611	14.470	11.703	10.230
RMSE [57]	4.8233	4.7275	4.0756	3.8039	3.4209	3.1984
NMI [58]	0.6065	0.5986	0.6983	0.9142	0.7536	0.7589
SF [59]	21.730	14.698	24.136	22.401	25.968	26.352
AG [60]	2.8956	2.1235	3.3210	2.9865	3.5620	3.5982

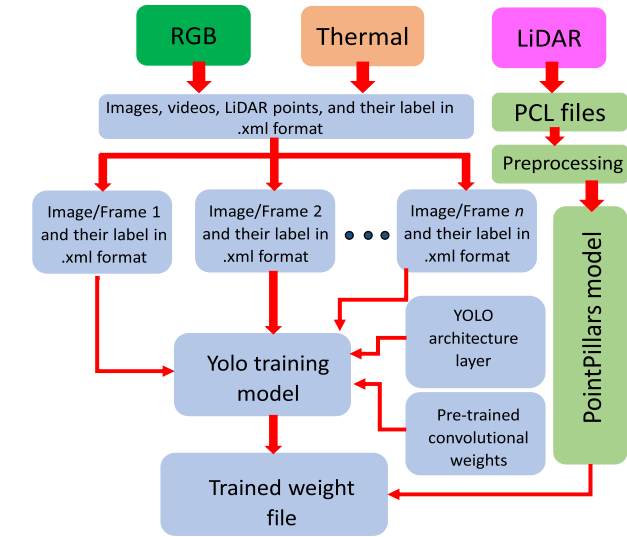


FIGURE 9. Shows the object detection model training using YoloV8 [35] and PointPillars [36] models.

We tested the detection model for single and multiple objects. The results showed a 93%-96% detection accuracy in real-time applications. Fig. 11 shows the comparison results of different state-of-the-art detection models. The YoloV8 [35] model has good detection results, as compared to the other four models RESNET-51 [38], SSD-50 [39], FRCN-50 [40], and YoloV5 [41]. We conducted experiments at different times to check the efficiency of the proposed model.

3) MODEL IMPLEMENTATION

The proposed deep learning-based multi-sensor fusion model is implemented on hardware and software (ROS) successfully. The deep learning sensor fusion model combines many forms of data to produce a holistic knowledge of a dataset. The proposed model fuses data from several sensor types and gets the best potential results. We can further increase our sensor fusion and detection model into an intelligent hybrid model, which can be a large powerful model to produce more stable results. We have evaluated our proposed models on three Datasets (FLIR, TarDAL, and CUHK). Tables 2, 3, and 4 illustrate the quantitative results of the proposed model and other sensor fusion models using eight different evaluation metrics. Figs. 9 and 10 show the qualitative results of our

proposed model and other models. Our model has the capacity to work on a large number of data points, which can make accurate predictions on unknown data.

We have tested our model in indoor and outdoor environments in Hong Kong (CUHK, HKPC, Hong Kong Science and Technology Park, Tai Wai, etc) to test its stability and validity. In real-time, our model is able to use cloud platforms like Google Cloud or Amazon Web Services for scalability and access to hardware resources. In the future, we can improve the model’s performance by employing techniques like regularization, hyperparameter tweaking, and feature selection. This ensures that the model is reliable and can forecast new data points in a timely and precise manner.

IV. EXPERIMENTAL RESULTS
A. PROPOSED AND PUBLIC DATASETS

The proposed dataset was collected from three sensors at the Chinese University of Hong Kong, the Hong Kong Productivity Council, and in other locations in Hong Kong, such as a footbridge at Tai Wai, Sham Shui Po Park, and the Hong Kong Science and Technology Park (HKSTP). This dataset was collected as a part of this study. It includes indoor and outdoor scenes. It is captured in normal and adverse weather conditions. The detail of each dataset is given below.

1) FLIR DATASET

The FLIR ADAS offers both thermal and visible spectrum frames that are annotated for the development of object recognition systems based on convolutional neural networks (CNNs) [47]. It contained 8000 thermal and RGB images and videos. This data was created to boost research on RGBT and to help the automotive industry to produce more reliable ADAS and autonomous vehicle systems. The size of each frame is 540 × 460. It contains more than 14 label categories (Car, Person, Bus, Truck, Bike, Motorcycle, Traffic Light, Dog, Street signs, Scooter, Fire Hydrant, Train, etc.).

2) TARDAL DATASET

It is the Dataset of the Dalian University of Technology [48]. It contains 8400 thermal and RGB images. It has 600 independent scenes with a resolution of 1024 × 768. A total of 34,407 labels have been manually tagged, with six different types of targets: people, cars, buses, motorcycles, lamps, and trucks. (Because of labor constraints, some targets may be mismatched or overlooked).

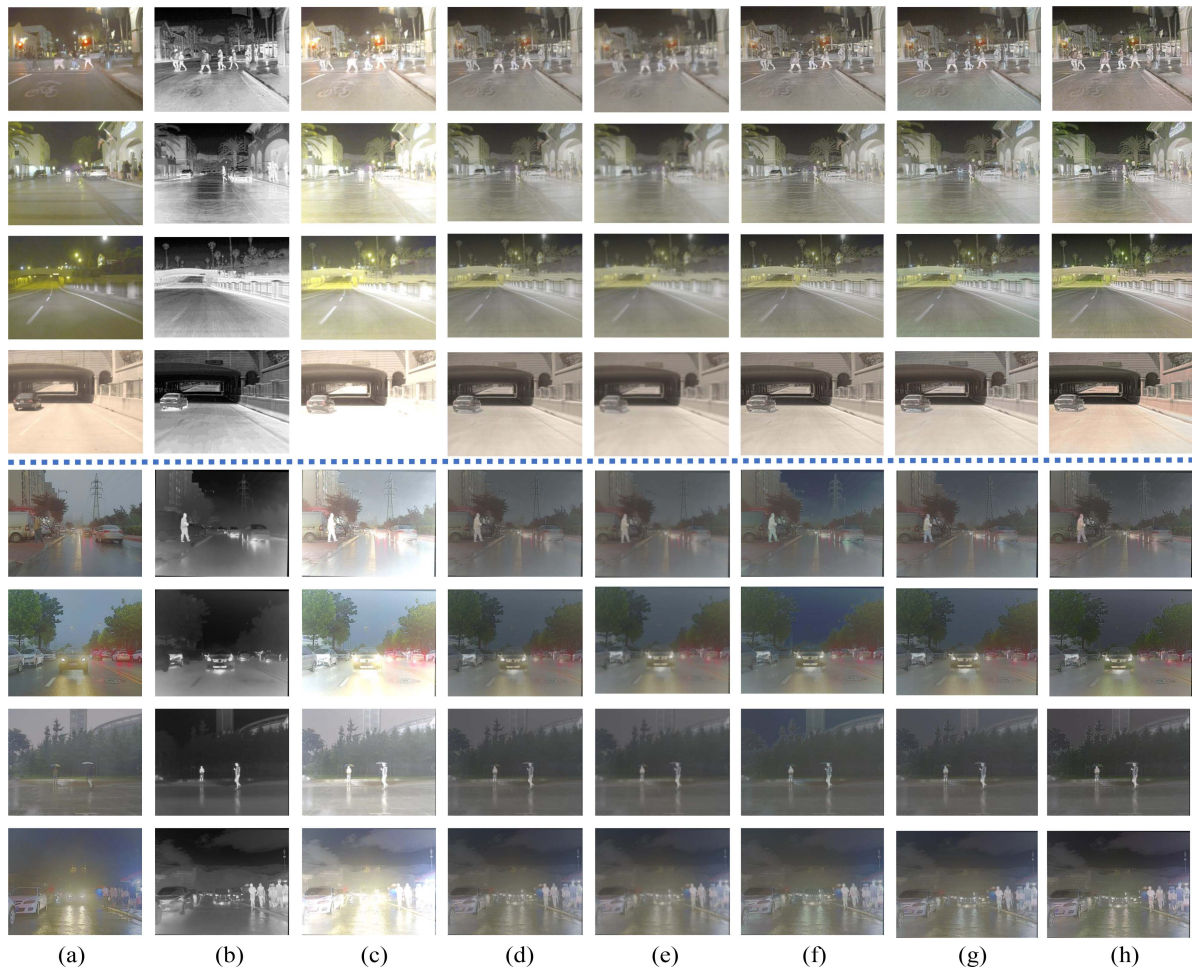


FIGURE 10. Illustrates the qualitative results of different fusion models. It contains images from two Datasets (the upper part FLIR dataset and the lower part TarDAL Dataset). (a) and (b) show the original RGB and Thermal frames, respectively. (c)-(h) show the result of the GFF [42] model, the MECNN [43] model, the LPSR [44] model, the MFCNN [45] model, MFIF [46] model, and (h) the proposed method (MSFF), respectively.

3) CUHK DATASET

It contains 59,116 RGB and Thermal frames, and 10,000 LiDAR frames in the form of 3D point clouds. All frames are labeled by experts using CVAT [49] and Roboflow [50] software. This Dataset has indoor and outdoor data from three different sensors RGB, Thermal, and LiDAR. It contains more than ten different label categories including the common road objects: Person, Bus, Truck, Bike, Traffic Light, Dog, Car, Fire Hydrant, Train, Scooter, Motorcycle, etc.

B. DATA LABELING AND TRAINING

We are using three types of software to label the data: Roboflow [51], CVAT(computer vision annotation tool) [52], and LiDAR labeler [53]. They took 2-4 months to label 120K images from the RGB and Thermal dataset.

1) LABELING

RGB and Thermal data labeling: We utilized Roboflow and CVAT for visual and thermal data annotation, respectively. Both are efficient and rapid labeling tools, featuring both automated and manual labeling capabilities, as well as a user-friendly graphical user interface (GUI) for annotating large

datasets. They employ the most prominent annotation formats, such as JSON, XML, CSV, and TXT, to annotate dozens of images quickly. It supports more than 15 formats for exporting data for training, both offline and online.

LiDAR data labeling: We produced comparable ground truth data and used the LiDAR Labeler software [53], a Matlab-based tool, to interactively label the ground truth images in a point cloud. This user-friendly app enables users to label their data in a 3D format and can import custom automation algorithms for automated labeling. It also provides the ground segmentation option to remove the empty area in the cloud points and allows users to suppress ground points in the point cloud, making it simpler to locate the objects. Additionally, this app can export labeled data into four different formats of LiDAR points (PCD, LAS, Velodyne, and Rosbag) for training purposes.

2) TRAINING

We have deployed a YOLOv8 and PointPillars-based AI detection model to detect objects in thermal, RGB, and LiDAR sensor videos. Due to the limited amount of publicly available

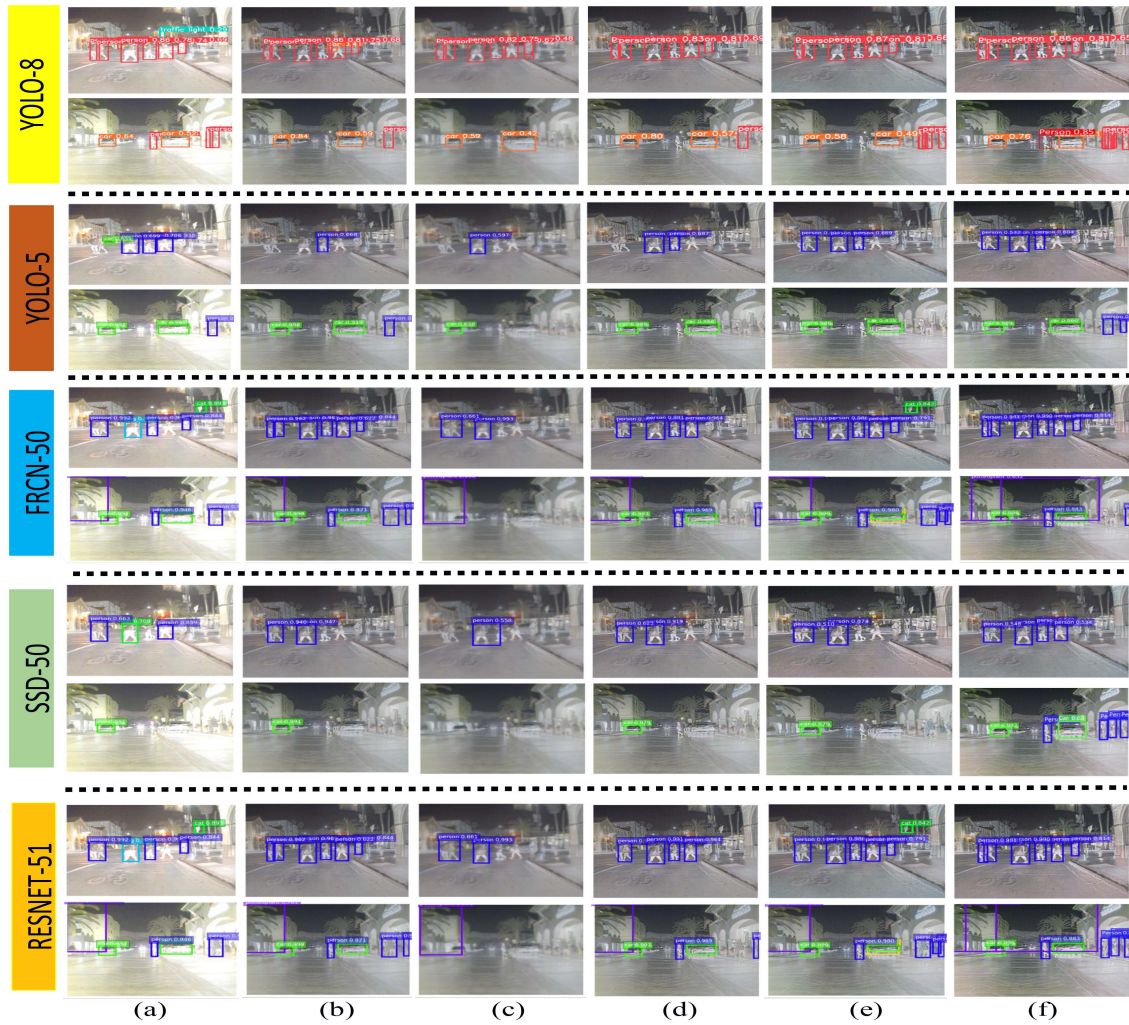


FIGURE 11. Shows the detection results using state-of-the-art detection models (Yolov8 [35], Yolov5 [41], FRCN-50 [40], SSD-50 [39], and RESNET-51 [38]) on fused images. (a)–(f) show the detection result on the GFF [42] model-based images, the MECNN [43] model-based images, the LPSR [44] model-based images, the MFCNN [45] model-based images, MFIF [46] fused images, and the proposed model, respectively.

data, we used data augmentation techniques to increase the dataset size. Below is a list of the deep learning models trained on each dataset.

a) Training of thermal data: We have trained a Yolov8 model on thermal Datasets with different classes (Car, Person, Bicycle, Dog, Traffic Signals, Bus, Truck, Motorcycle, Street Lights, etc.) and scenes (Day, Afternoon, Evening, Night, Rain, etc.). This model has been trained on 59,116 thermal frames, with a frame rate of 30 frames per second. The thermal dataset is separated into three parts: 80% for training, 10% for validation, and 10% for testing. After training the model for 500 to 1,000 epochs, the testing accuracy of this dataset ranges from 90% to 96% on average.

b) Training of RGB data: We trained the Yolov8 model on a Visual Dataset consisting of 59,116 RGB frames from various scenes (Day, afternoon, evening, night, rain, etc.) with various classes (Car, Person, Bicycle, Dog, Traffic signals, Bus, Truck, Motorcycle, Street lights, etc.). A total of 59,116 images were used for training. The dataset was divided as

follows: 80% for training, 10% for validation, and 10% for testing. Our model was trained using this dataset over 500 to 1,000 epochs, achieving an average testing accuracy of approximately 93–96%.

c) Training of LiDAR data: We trained the PointPillars model [54] on a LiDAR Dataset consisting of 10,000 LiDAR frames from various scenes (Day, afternoon, evening, night, rain, etc.) with various classes (Car, Person, Bicycle, Dog, Traffic signals, Bus, Truck, Motorcycle, Street lights, etc.). The data were divided into three parts: 80% for training, 10% for validation, and 10% for testing. It is trained with 500 and 1,000 epochs and yielded an average testing accuracy of 90%.

C. RESULTS EVALUATION

In this section, we have compared the qualitative and quantitative results of the proposed model with other state-of-the-art fusion models, as shown in Tables 2, 3, and 4. These results provide sufficient information to assess their performance. To objectively measure the performance, we have used eight

commonly used metrics on FLIR, TarDAL, and CUHK image datasets. These metrics are Visual Information Fidelity (VIF) [55], Mean Square Error (MSE) [57], Normalized Mutual Information (NMI) [58], Root Mean Square Error (RMSE) [57], Spatial Frequency (SF) [59], and Average Gradient (AG) [60]. These metrics calculate the visual information of the fused image by comparing it with the original images. They measure the amount of information retained from the two source images in the fused image and assess the structure of the combined image. These metrics help us to understand the algorithms' ability to fuse visual information, structural information, and image details. For this reason, these eight metrics are appropriate to use.

1) VISUAL INFORMATION FIDELITY (VIF)

It aims to simulate the psycho-visual and physiological aspects of the visual system that are not present in the human visual system [61]. This technology is designed for natural images that people typically view; it hasn't been thoroughly tested with other types of images, such as graphics or medical images. VIF [55] usually ranges between 0 and 1, but it can exceed 1 in some cases when an image is considered to have a higher resolution [62].

$$VIF = \frac{\sum_{j \in \text{subband}} I(C^{N,J}; F^{N,J} | S^{N,J})}{\sum_{j \in \text{subband}} I(C^{N,J}; E^{N,J} | S^{N,J})} \quad (19)$$

$$I(C^{N,J}; F^{N,J} | S^{N,J}) = \frac{1}{2} \sum_{x=1}^W \sum_{y=1}^H \log_2 \left(1 + \frac{g_x^2 s_x^2 \lambda_x}{\sigma_v^2 + \sigma_n^2} \right) \quad (20)$$

$$I(C^{N,J}; E^{N,J} | S^{N,J}) = \frac{1}{2} \sum_{x=1}^W \sum_{y=1}^H \log_2 \left(1 + \frac{s_x^2 \lambda_x}{\sigma_n^2} \right) \quad (21)$$

where $I(C^{N,J}; F^{N,J} | S^{N,J})$ denote the information that the fused image would optimally retrieve from the certain band in the RGB and Thermal images. Furthermore, let y represent an eigenvalue, s_x the realization for a matching reference image, g_x^2 the attenuation factors for a particular reference image, and σ_n^2 the variance of visual noise.

2) STRUCTURAL SIMILARITY INDEX (SSIM)

According to Wang et al. [56], [63], SSIM is a newer measurement tool compared to PSNR. SSIM has three basic properties: contrast, local average brightness, and structural information, which is distinct from both brightness and contrast. This image quality metric is based on the human visual system (HVS). Additionally, SSIM is considered to be more responsive to histogram analysis when evaluating different key steganographic techniques.

$$SSIM(I, J) = \frac{(2\mu_1 \hat{\mu}_1 + C_2)}{(\mu_1^2 + \hat{\mu}_1^2 + C_1)(\sigma_1^2 + \hat{\sigma}_1^2 + C_2)} \quad (22)$$

$$C_1 = (s_{c1}D)^2 \quad (23)$$

$$C_2 = (s_{c2}D)^2 \quad (24)$$

where μ shows the mean intensity value of image, σ shows the deviation value. The c_1 and c_2 are stabilizing parameters. D shows the dynamic values of pixels in RGB and Thermal images.

3) MEAN STRUCTURAL SIMILARITY INDEX (MSSIM)

This method measures the structural similarity index across the entire image [57]. The average value of the local image window is used to compute the MSSIM [63].

$$MSSIM(I, J) = \frac{1}{M} \sum_{x=1}^M SSIM(I_x, \hat{I}_x) \quad (25)$$

M represents the number of local windows in the image. The local windows of the images are denoted by I_x and $\hat{I}_x$, where x indicates the window index.

4) MEAN SQUARE ERROR (MSE)

The Mean Squared Error (MSE) [57] is a method used to determine the degree of error in image pixels by comparing them to a reference. It is calculated as:

$$MSE = \frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H (I - J)^2 \quad (26)$$

where W and H are the width and height of the images. x and y are image coordinates.

5) ROOT MEAN SQUARE ERROR (RMSE)

Root Mean Squared Error (RMSE) [57] is a variant of Mean Squared Error (MSE) where a square-root operation is performed, resulting in a lower value, but representing the same overall error as the MSE.

$$RMSE = \sqrt{\frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H \sum_{z=1}^L (I - J)^2} \quad (27)$$

6) NORMALIZED MUTUAL INFORMATION (NMI)

It is used to measure the mutual information between the original information and the fused images [58]. Suppose we have image A (RGB), image B (Thermal), and fused image f ; then we calculate the NMI as:

$$H_{AB}(a, b) = \sum_{n=1}^N \sum_{m=1}^M H_{AB}(I_A(m, n), I_B(m, n)) \quad (28)$$

$$P_{AB}(a, b) = \frac{1}{2NM} * H_{AB}(a, b) \quad (29)$$

$$I_A(f, a) = \sum_{f,a} P_A(f, a) \log \frac{P_A(f, a)}{P_f(f)P_A(a)} \quad (30)$$

$$I_B(f, b) = \sum_{f,b} P_B(f, b) \log \frac{P_B(f, b)}{P_f(f)P_B(b)} \quad (31)$$

$$M_{AB}^f = I_A(f, a) + I_B(f, b) \quad (32)$$

where H_{AB} is mutual entropy, P_A and P_B are the entropies of images A , and B , respectively. M_{AB}^f is normalized mutual information of images A , B and f .

7) SPATIAL FREQUENCY (SF)

It quantifies the total activity level of an image [59]. Due to the complexity of the human visual system, it is challenging to calculate an image visual quality. Therefore, the formula for the Spatial Frequency (SF) is as follows:

$$RF = \sqrt{\frac{1}{MN} \sum_{m=1}^{M-1} \sum_{n=1}^{N-1} [F(m, n) - F(m, n-1)]^2} \quad (33)$$

where F is the gray value of the image at position (m, n) .

$$CF = \sqrt{\frac{1}{MN} \sum_{m=1}^{M-1} \sum_{n=1}^{N-1} [F(m, n) - F(m-1, n)]^2} \quad (34)$$

$$SF = \sqrt{(RF)^2 + (CF)^2} \quad (35)$$

where SF is the total spatial frequency of the image obtained by row frequency (RF) and column frequency (CF).

8) AVERAGE GRADIENT (AG)

It measures the edge detail of the image [60]. It is the average of the horizontal and vertical gradient values of an image. It is calculated as:

$$K = \frac{1}{4} \times \frac{1}{(M-1)(N-1)} \quad (36)$$

$$AG = K \times \sum_{m=1}^{M-1} \sum_{n=1}^{N-1} \sqrt{\left(\frac{\delta I(m, n)}{\delta m}\right)^2 + \left(\frac{\delta I(m, n)}{\delta n}\right)^2} \quad (37)$$

where $\frac{\delta I}{\delta m}$, and $\frac{\delta I}{\delta n}$ are horizontal and vertical gradient values in the image at (m, n) coordinates.

V. DISCUSSION

A. COMPARISON WITH OTHER CNN MODELS

In this section, we compare our proposed method with state-of-the-art fusion models. Fig. 10 shows that the visual quality of MSFF outperforms the GFF [42], MECNN [43], LPSR [44], MFCNN [45], and MFIF [46] models. The last column shows the results of our proposed model. In terms of visual information and the similarity index, our MSFF method outperforms the GFF [42], MECNN [43], LPSR [44], MFCNN [45], and MFIF [46] models. Furthermore, the LPSR [44] model outperforms the GFF [42], and MECNN [43] models in terms of generalization. The creation of sensor fusion deep learning models for autonomous cars is an active subject of study, with several intriguing options for future advancement [64], [65]. Sensor fusion deep learning models are projected to become more practical for application in autonomous cars as deep learning architectures continue to progress. The integration of unstructured and changing data sources is one of the main novel areas that will impact

the future development of sensor fusion deep learning models in autonomous cars. This method will allow deep neural networks to learn the contextual awareness and reasoning required to make autonomous judgments in an unstructured and dynamic environment.

Another significant area of progress in the coming years will be the incorporation of federated learning [66]. Unlike standard deep learning models, which are trained on centralized, controlled Datasets, federated learning will allow autonomous cars to make judgments based on local data while also relying on large global information. This distributed method will assist autonomous cars in learning from experience and adapting to local settings more quickly and accurately.

Convolutional neural networks (CNNs)-based sensor fusion models were tested on different AVs. In terms of accuracy and overall generalization performance, the CNN model has proven to be much better than the machine learning-based models. Furthermore, the CNN model outperformed all other models examined in terms of sensor fusion similarity. Overall, the study's findings indicate that CNNs are best suited for sensor fusion in autonomous cars.

B. STABILITY

When paired with flawless perception and motion planning, sensor fusion deep learning models increase the accuracy and stability of autonomous cars. Deep learning algorithms are used to integrate data from multiple sensors, including cameras, LiDAR, IMUs, and radars, in order to properly locate autonomous cars in their surroundings and recognize things around them. Sensor fusion also aids in reducing the computing complexity, or amount of labor required, for autonomous cars to plan and operate in their surroundings. Deep learning sensor fusion models are also resistant to sensor noise and changes in the environment [67]. Deep learning's generalization ability allows it to avoid becoming over-specialized in the source dataset and to effectively apply it to new instances in the environment, such as a vehicle traveling in a different season, at a different time, or in a different area. This resilience improves the stability and dependability of autonomous vehicles' navigation and behavior judgments even when the environment changes slightly.

In general, deep learning models are capable of properly fusing data from several sensors, allowing the vehicle to correctly locate and recognize objects in its environment. Deep learning algorithms are also resistant to sensor noise and environmental changes, allowing autonomous cars to continue navigating even when minor changes occur. Sensor fusion deep learning models will grow more stable and accurate over time as more dependable and diverse datasets for training become available.

C. COMPUTATIONAL TIME

The complexity of sensor fusion deep learning models in autonomous cars is rising [68]. This growth in complexity is due to their necessity to interpret and analyze more diverse

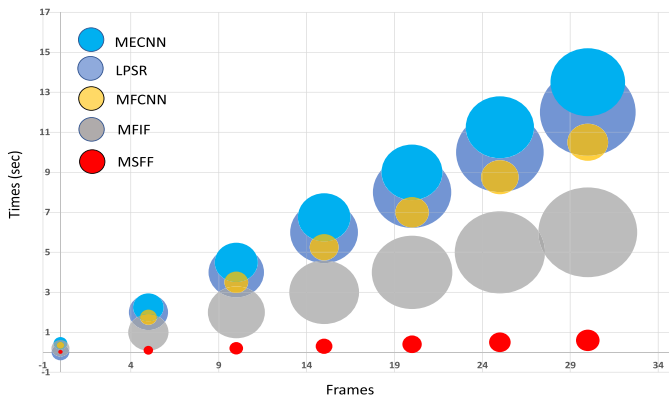


FIGURE 12. Shows the computational cost for different models (best view in color). The red color represents the proposed model (MSFF).

data sources in order to make optimal judgments. In short, the computational cost associated with these models is quite high and rapidly increasing.

As shown in Fig. 12, our method can produce a 520×520 fusion image from two input images in about 0.02 seconds on a system with an NVIDIA RTX3090 GPU and an Intel Core i7-3770K CPU. In comparison, the moderate form of MFCNN takes approximately 0.33 seconds to fuse two 520×520 input images. Using MECNN, it takes around 0.07 seconds to fuse two 512×384 input images on an Intel Xeon processor. Additionally, GFF takes 0.33 seconds, and LSPR takes 0.25 seconds. These results demonstrate that our proposed solution is computationally efficient.

According to Xilinx Research Labs [69], the computing cost per frame of sensor fusion deep learning is heavily reliant on the size of the sensor fusion network, regardless of the Autonomous Vehicle's Hardware platform or methodology. Modern autonomous cars use advanced CPU architectures and modular processors to manage higher costs. Examples include GPU accelerators for sensor fusion workloads, FPGAs, and application-specific integrated circuits (ASICs).

VI. CONCLUSION

We proposed a comprehensive approach to build a robust cognitive system for AV by integrating thermal sensors with RGB and LiDAR sensors to address current challenges in object detection in adverse weather conditions. The proposed deep learning-based multi-sensor fusion technique consists of visual information fusion and object detection models. A convolutional neural network is used to develop a fusion framework. Meanwhile, the object detection framework uses the latest version of the YoloV8 model, which shows high accuracy in real-time detection. We used an advanced annotation tool to label the data (images and cloud points). We tested our fusion and detection models in adverse weather conditions, such as rain, fog, smog, smoke, and haze, and found them to work efficiently. In the future, we can further improve the fusion framework by applying suitable data calibration

techniques that support varying FOVs, data formats, and data compatibility.

Conflict of interest: All authors have no conflict of interest

Data availability: All data are available on reasonable request.

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